Weak function extensionality implies the full function extensionality axiom

The Univalent Foundations Program

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Editorial note. The counted statement is the author's labelled theorem wfetofe at source lines 980-983; its proof environment starts on line 985, the line immediately after the blank line that follows the statement environment, so the pairing is by adjacency and the retained proof range 985-1007 is the whole environment.

Editorial note. The context span at source lines 975-978 is the author's own introduction to the theorem and is retained because the proof refers back to the canonical map introduced there; it carries no counter value.

Editorial note. The two numbered displays inside the retained proof are the author's own labelled equations of the proof; because the book lets the equation counter share the theorem counter, the retained counters make the counted theorem print as Theorem 4.9.5 and those displays print as (4.9.6) and (4.9.7), exactly as in the book.

Editorial note. This source directory intentionally contains only the author file, so the retained book macro library is bundled as a per-source copy at sources/hott-equivalences/retained-preamble.tex, referenced from preamble_spans; the copy is verbatim from the same pinned book snapshot and no macro is added, changed or invented.

Editorial scope. One original result of the HoTT Book, counted once: weak function extensionality implies the function extensionality axiom (author label wfetofe, original Theorem 4.9.5).

Allowed standard prerequisites (editorial).

- The map from identities to homotopies. There is a canonical map sending an identity of functions to the pointwise homotopy it induces.. It is the map whose being an equivalence is the conclusion, used at source lines 988 and 994.
- Equivalences and fibers. A map is an equivalence exactly when all its fibers are contractible, and a fiberwise map induces an equivalence on total spaces exactly when it is fiberwise an equivalence.. It is the reduction performed at source line 990 and the contractibility criterion used at source lines 995-999.
- Weak function extensionality. A product of contractible types over any index type is contractible.. It is the hypothesis applied at source line 1006.

- Retracts. If a type is a retract of a contractible type then it is contractible.. It is the step taken at source lines 1004-1006.

Source reference. Original theorem 2.9.3: sources/hott-equivalences/book-labelnumbers-first-edition.tex, source lines 157–157. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Source reference. Original equation 2.9.2: sources/hott-equivalences/book-labelnumbers-first-edition.tex, source lines 1256–1256. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Source reference. Original theorem 3.11.8: sources/hott-equivalences/book-labelnumbers-first-edition.tex, source lines 1760–1760. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Source reference. Original theorem 4.7.7: sources/hott-equivalences/book-labelnumbers-first-edition.tex, source lines 1891–1891. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Source reference. Original theorem 2.15.7: sources/hott-equivalences/book-labelnumbers-first-edition.tex, source lines 1909–1909. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

We now show that weak function extensionality implies the usual function extensionality. Recall from (2.9.2) the function $\mathsf{happly}(f,g) : (f = g) \to (f \sim g)$ which converts equality of functions to homotopy. In the proof that follows, the univalence axiom is not used.

Theorem 4.9.5. *Weak function extensionality implies the function extensionality original theorem 2.9.3.*

Proof. We want to show that

$$\prod_{(A:\mathcal{U})} \prod_{(P:A\to\mathcal{U})} \prod_{(f,g:\prod_{(x:A)} P(x))} \mathsf{isequiv}(\mathsf{happly}(f,g)).$$

Since a fiberwise map induces an equivalence on total spaces if and only if it is fiberwise an equivalence by original theorem 4.7.7, it suffices to show that the

function of type

$$\Big(\sum_{g:\prod_{(x:A)} P(x)} (f = g)\Big) \to \sum_{g:\prod_{(x:A)} P(x)} (f \sim g)$$

induced by $\lambda(g : \prod_{(x:A)} P(x)).\, \mathsf{happly}(f, g)$ is an equivalence. Since the type on the left is contractible by original theorem 3.11.8, it suffices to show that the type on the right:

$$\sum_{(g:\prod_{(x:A)} P(x))} \prod_{(x:A)} f(x) = g(x) \tag{4.9.6}$$

is contractible. Now original theorem 2.15.7 says that this is equivalent to

$$\prod_{(x:A)} \sum_{(u:P(x))} f(x) = u. \tag{4.9.7}$$

The proof of original theorem 2.15.7 uses function extensionality, but only for one of the composites. Thus, without assuming function extensionality, we can conclude that (4.9.6) is a retract of (4.9.7). And (4.9.7) is a product of contractible types, which is contractible by the weak function extensionality principle; hence (4.9.6) is also contractible. □